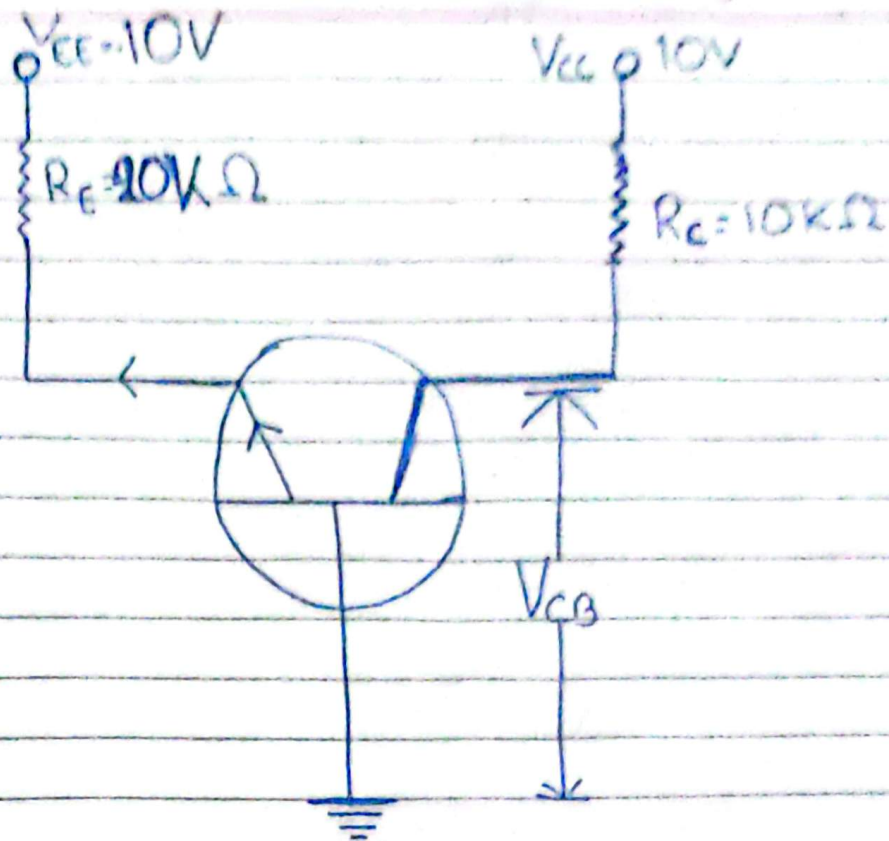




Draw the DC load line

Solⁿ:

$$V_{CB} = V_{CC} - I_C R_C \quad [\text{From } V_{CC} = I_C R_C + V_{CB}]$$

or $I_C R_C = V_{CC} - V_{CB}$

or $I_C = \frac{V_{CC} - V_{CB}}{R_C} \dots \dots \textcircled{i}$

substituting

$$I_C = \frac{10 - V_{CB}}{10000}$$

Now,

Date _____
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At ~~(saturation)~~ Cut-off point:

$$I_c = 0$$

$$\text{or, } \frac{10 - V_{CB}}{10000} = 0$$

$$\text{or, } 10 - V_{CB} = 0$$

$$\text{or, } V_{CB} = 10V$$

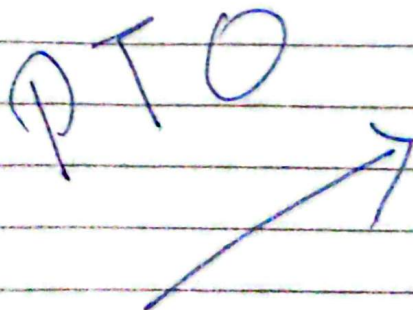
At Saturation point:

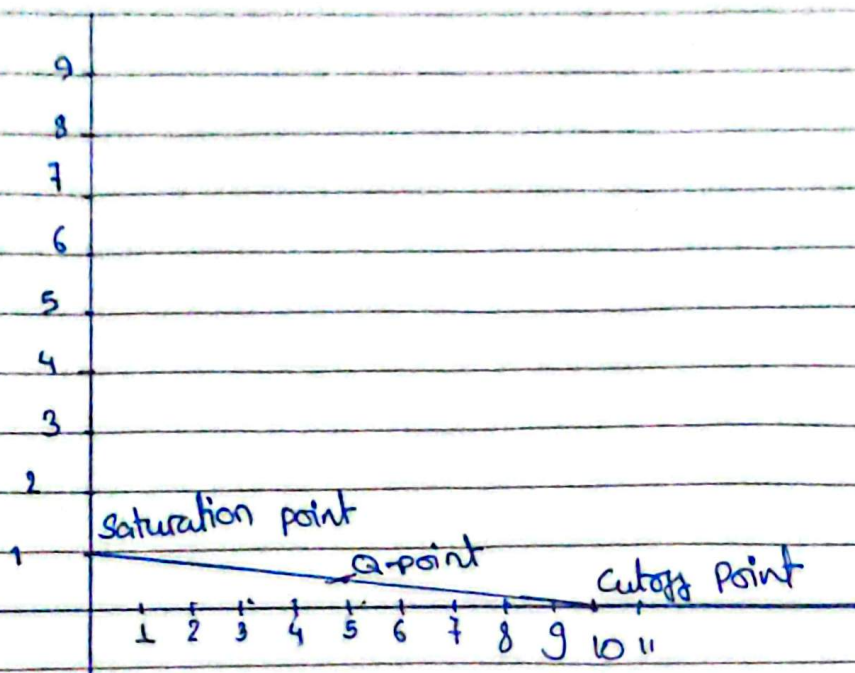
$$V_{CB} = 0$$

$$\text{or, } I_c = \frac{10 - 0}{10000}$$

$$\text{or, } I_c = 1mA$$

$\therefore$ Cutoff point = $(10V, 0)$ and Saturation point = $(0, 1mA)$





For Q-point

$$V_{EE} = I_E R_E$$

$$I_E = \frac{V_{EE}}{R_E}$$

$$I_E = \frac{10}{20 \times 10^3}$$

$$I_E = 0.5 \text{ mA}$$

Again

$$V_{CB} = V_{CC} - I_C R_C$$

$$V_{CB} = 10 - 0.5 \times 10^{-3} \times 10 \times 10^3$$

$$V_{CB} = 10 - 5$$

$$V_{CB} = 5 \text{ V}$$

$$\begin{aligned} \text{Q-point} &= (I_E, V_{CB}) \\ &= \left(\frac{I_C}{2}, \frac{V_{CC}}{2} \right) \\ &= (0.5 \text{ mA}, 5 \text{ V}) \end{aligned}$$

